

Macroeconomic Theory and Policy: Lecture 2 (Ch.2)

What are You Going to Learn (ch. 2)

- The importance of gross domestic product (GDP)
- The composition of GDP, and how it has changed over time
- How to use GDP to examine
 - ▶ The evolution of living standards
 - ▶ Differences in living standards across countries

Introduction: When did the Measurement of GDP Begin?

- How severe was the Great Depression? Policymakers in the 1930s did not have current data measures
- Stock prices, railroad freight reports, and some limited measures of industrial production did signal problems, but there was no broad-based measure of economic activity available
- Policymakers were unable to quantify the Depression or to indicate the effectiveness of policies designed to jumpstart the economic recovery
- This event motivated the U.S. Department of Commerce to create the National Income and Product Accounts to provide an overview of the state of the economy

Measuring the State of the Economy

Gross domestic product (GDP): The market value of the final goods and services produced in an economy over a certain period

- Production measure of GDP: The number of goods produced in the economy
- Expenditure measure of GDP: The total purchases in the economy
- Income measure of GDP: All the income earned in the economy
- All three approaches give identical measures of GDP:

$$\text{Production} = \text{Expenditures} = \text{Income}$$

- ▶ This equation should hold in the ideal condition, but, in practice, measurement error, among other factors, cause these measures to be slightly different

The Expenditure Approach to GDP

- The national income accounting identity states:

$$Y = C + I + G + NX$$

where

- ▶ $Y = \text{GDP}$
 - ▶ $C = \text{consumption}$
 - ▶ $I = \text{Investment}$
 - ▶ $G = \text{Government purchases}$
 - ▶ $NX = X - M = \text{Export} - \text{Imports}$
- This is probably one of the most important identity in Macroeconomics
 - Only final good and service would be counted

Investment

- Business fixed investment (non-residential)
 - ▶ Spending by firms on plants, machinery and equipment
- Residential investment
 - ▶ Construction of new houses and apartment buildings
 - ▶ However, rent and old home resales enter into GDP through consumption as housing services produced
 - ▶ Even if you own your house and don't pay rent, you still receive "housing services." Therefore, the value of housing imputed (estimated): what you would pay if you rented.

Investment

- Inventory investment

- ▶ Changes in inventories (of final or intermediate goods)
- ▶ Inventories are goods that are produced but not yet sold
 - ★ When there is an increase in inventories, this is treated as if the firm purchases these goods from themselves and are included in GDP
 - ★ Later, when inventories are sold as final goods, this results in a decrease in inventories (GDP) and an increase in consumption (also included in GDP)

Example of a National Account, US 2018

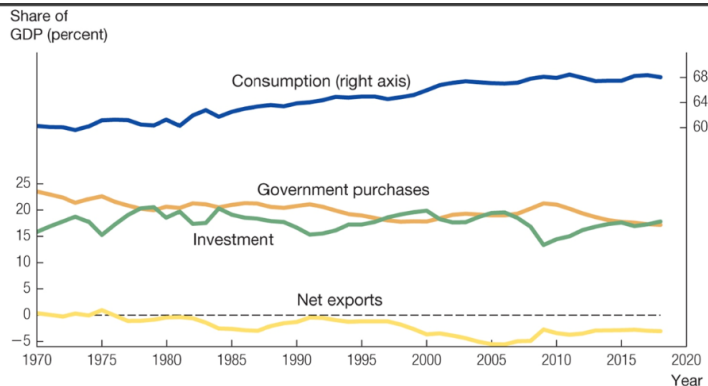
Variables	2018	% of GDP	Per Capita
Gross Domestic Product	20,580.20	100.00	62,794.60
Personal Consumption Expenditures	13,998.70	68.02	42,713.03
Goods	4,364.80	21.21	13,317.94
Durable Goods	1,475.60	7.17	4,502.37
Nondurable goods	2,889.20	14.04	8,815.57
Services	9,633.90	46.81	29,395.09
Gross Private Domestic Investment	3,628.30	17.63	11,070.72
Fixed Investment	3,573.60	17.36	10,903.82
Nonresidential	2,786.90	13.54	8,503.43
Structures	633.2	3.08	1,932.03
Equipment	1,222.60	5.94	3,730.41
Intellectual Property Products	931.1	4.52	2,840.99
Residential	786.7	3.82	2,400.39
Change in private inventories	54.7	0.27	166.90
Net Exports of Goods and Services	-638.2	-3.10	-1,947.28
Exports	2,510.30	12.20	7,659.46
Goods	1,661.30	8.07	5,068.98
Services	848.9	4.12	2,590.18
Imports	3,148.50	15.30	9,606.75
Goods	2,570.60	12.49	7,843.45
Services	577.9	2.81	1,763.30
Government Consumption Expenditures and Gross Investment	3,591.50	17.45	10,958.44
Federal	1,347.30	6.55	4,110.90
National Defense	793.6	3.86	2,421.44
Nondefense	553.7	2.69	1,689.46
State and Local	2,244.20	10.90	6,847.54

Source: Bureau of Economic Analysis

Notes: Billions of Dollars (per capita in dollars): Last Revised on: September 30, 2019

Expenditure Shares of U.S. GDP

- The composition of GDP has been relatively stable over time
- Consumption has increased as a share of GDP recently
 - ▶ NX being significantly negative
 - ▶ Decline in government purchases



Source: U.S. Department of Commerce, Bureau of Economic Analysis, www.bea.gov.
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The Income Approach to GDP

- The income approach: Measures the sum of all income earned in the economy - from both labor and capital
- For every dollar of product sold there is a dollar of income earned
- Capital:
 - ▶ Inputs into production other than labor that are not used up in the production process
 - ★ Consider the cafe as an example: the coffee beans are not considered capital, but the coffee machine is regarded as a capital asset
 - ▶ Increased by firms through investment
- Depreciation: The deterioration of the capital stock due to wear and tear

$$\text{GDP} - \text{depreciation} = \text{net domestic product}$$

The Income Approach to U.S. GDP in 2018

- The majority of income (53 percent) comes from compensation to employees
- The next largest category is the net operating surplus of businesses – “profits” earned by business owners, which provide both labour and capital

TABLE 2.2

The Income Approach to U.S. GDP in 2018

	Total (billions of dollars)	Share of GDP (percent)	Per person (dollars)
Gross domestic product	20,500	100.0	62,600
Compensation of employees	10,870	53.0	33,190
Wages and salaries	8,850	43.2	27,020
Benefits	2,020	9.9	6,170
Taxes less subsidies on businesses	1,360	6.6	4,160
Net operating surplus of businesses	5,900	28.8	18,000
Depreciation of fixed capital	3,270	16.0	10,000

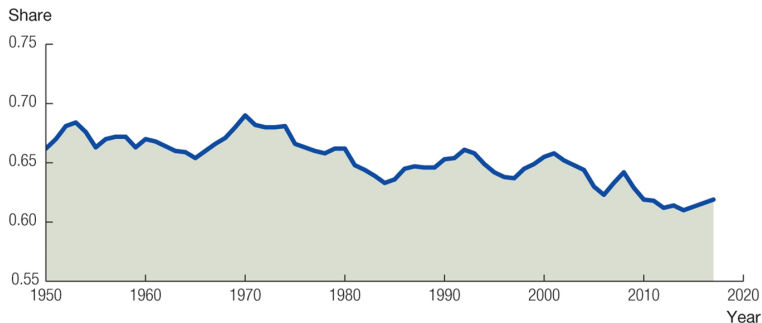
Source: U.S. Department of Commerce, Bureau of Economic Analysis, www.bea.gov.
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Total Share of GDP to Inputs

- A convenient way to present this number is by splitting income into labour income and capital income
- After adjusting the mixed income (income that involve both labour and capital),
 - ▶ Share of GDP to labour: Two-thirds (approx.)
 - ▶ Share of GDP to capital: One-third (approx.)

Labor's Share of GDP

- Notice that labor's share of income fluctuates very little over time: between 60 and 70 percent of GDP
- There is no real general trend to labor's share of income, but maybe we are the tipping point of falling in labour share
 - ▶ In other words, it is not the case that capital owners are becoming richer at the expense of labor, or vice versa



Source: U.S. Department of Commerce and author's calculations.
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The Production Approach to GDP

- Recall that GDP is the value of final goods and services produced in an economy during a given period
 - ▶ Do not want “double counting” in GDP
 - ▶ If a steel company produces \$10 million worth of steel and that steel is then used by an automobile company to make \$100 million worth of cars:
 - ★ The value of the steel should not be counted twice
 - ★ GDP goes up by only \$100 million (the value of the final car sale)
- Value added
 - ▶ The amount each producer contributes to GDP
 - ▶ The revenue generated by each producer minus the value of intermediate products
- Only new production of goods and services counts toward GDP

A Hypothetical Case

	AT Cotton	ST Fabric	Savage Clothing	Total Factor Income
Value of sales	3500	8000	16000	
Intermediate goods		3500	8000	
Wages	1500	2000	2500	6000
Interest	200	400	600	1200
Rent	800	1200	1600	3600
Profit	1000	900	3300	5200
Total expenditure	3500	8000	16000	
Value added	3500	4500	8000	

- For this simple economy, the GDP can be calculated in three approaches
- Summing the all the red cell give the income approach GDP, green give the value added approach, and yellow gives the expenditure approach

What Is Included in GDP and What Is Not?

- Included:

- ▶ Government spending on goods/services
- ▶ Factory production
- ▶ Healthcare expenditures
- ▶ Ingredients and food purchased
- ▶ Kids in day care

- Not included:

- ▶ Government transfer payments
- ▶ Environmental conditions
- ▶ A measure of a nation's health
 - ★ See: <https://www.gapminder.org/videos/200-years-that-changed-the-world/>
- ▶ Time spent cooking at home
- ▶ Babysitter

Beyond GDP

- GDP is used by economists as a proxy for standards of living, but economist want to create broader measure
- Jones and Klenow (2016) started with looking at consumption per person instead of production
- Motivated by the typical building board in macroeconomics and in microeconomics, they also incorporate life expectancy, leisure and consumption inequality in to the analysis

Beyond GDP: GDP and Welfare across Countries in Year 2007

Country	Per capita GDP	Welfare	Decomposition			
			Life Expectancy	C/Y	Leisure	Inequality
US	100	100	0	0	0	0
			77.8	0.845	836	0.658
Sweden	79.4	91.2	0.181	-0.186	0.01	0.135
			80.9	0.701	807	0.404
Japan	71.3	82.6	0.265	-0.154	-0.028	0.063
			82.5	0.724	912	0.554
South Korea	58.3	45.3	0.078	-0.29	-0.116	0.076
			79.3	0.632	1120	0.531
Vietnam	5.9	4	-0.082	-0.269	-0.02	-0.006
			74.2	0.645	893	0.668
Kenya	2.8	1.9	-0.394	0.104	0.059	-0.157
			54.4	0.938	644	0.865

The US's per capita GDP and welfare are normalized to 100, while the other are normalized to 0

Measuring Changes Over Time

- When examining GDP over time, we need to take into account changes in prices
- Nominal GDP: A measure of GDP when prices and quantities have not been separated, using current year prices
 - ▶ $\text{Nominal GDP} = \text{price level} \times \text{Real GDP}$
 - ▶ If prices are increasing over time, then GDP will increase, regardless of whether or not there is an increase in quantity.
- Real GDP: Actual quantity of goods and services, using base year prices
 - ▶ $\text{Real GDP} = \frac{\text{Nominal GDP}}{\text{Price Level}}$

Measuring Changes Over Time

- To compute GDP across time, we must use a certain year's prices.
 - ▶ Real GDP will be measured in a certain year's dollars
 - ▶ Nominal GDP is measured in current dollars
- Think of there are N different good/service in the economy, each of them have their own price
- Nominal GDP: $Y_t = P_{1,t}Q_{1,t} + P_{2,t}Q_{2,t} + \cdots + P_{N,t}Q_{N,t}$ where N is the total number of goods and services in the economy and $P_{i,t}$ are prices of good i in year t
- Real GDP: $Y_t = P_{1,b}Q_{1,t} + P_{2,b}Q_{2,t} + \cdots + P_{N,b}Q_{N,t}$ where $P_{i,b}$ is the price of good i in the base year

An Example of Changes in Nominal GDP

- Consider an economy that produces two goods: donuts (d) and pizza (p).
- Nominal GDP in 2020:

$$Y_{2020} = P_{d,2020} Q_{d,2020} + P_{p,2020} Q_{p,2020}$$

- Real GDP in 2020 (base year = 2014):

$$Y_{2020}^R = P_{d,2014} Q_{d,2020} + P_{p,2014} Q_{p,2020}$$

- If the quantity of goods and services produced does not change, but prices do change:
 - ▶ Nominal GDP will change
 - ▶ Real GDP will not change

An Example of Changes in Nominal GDP

Year	Price of Donuts	Quantity of Donuts	Price of Pizza	Quantity of Pizza
2014 (base)	10	50	25	100
2015	12	50	28	100

- Nominal GDP in 2014 = $10 \times 50 + 25 \times 100 = 3000$
- Nominal GDP in 2015 = $12 \times 50 + 28 \times 100 = 3225$
- Real GDP in 2015 = $10 \times 50 + 25 \times 100 = 3000$
- The real GDP in 2015 is the same as the nominal GDP in 2014, because only the price has increased

An Example of Changes in Nominal GDP

Year	Price of Donuts	Quantity of Donuts	Price of Pizza	Quantity of Pizza
2014 (base)	10	50	25	100
2015	12	55	28	105

- Nominal GDP in 2015 = $12 \times 55 + 28 \times 105 = 3660$
- Real GDP in 2015 = $10 \times 55 + 25 \times 105 = 3175$

Another Calculation

TABLE 2.4

Real and Nominal GDP in a Simple Economy, 2025–2027

	2025	2026	2027	Percentage change 2026–2027
Quantity of apples	500	500	550	10.0
Quantity of computers	5	5	6	20.0
Price of apples (dollars)	1	2	3	50.0
Price of computers (dollars)	900	1,000	1,000	0.0
Nominal GDP	5,000	6,000	7,650	27.5
Real GDP in 2025 prices	5,000	5,000	?	?
Real GDP in 2026 prices	6,000	6,000	7,100	18.3
Real GDP in 2027 prices	?	6,500	7,650	17.7
Real GDP in chained prices, benchmarked to 2027	?	6,483	7,650	18.0

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Quantity Indexes

Calculating real GDP changes over time:

- The Laspeyres index: Calculates changes in real GDP using the initial prices
 - ▶ e.g., for year 2026 to 2027,

$$\text{Laspeyres index} = \frac{7100}{6000} = 1.183$$

- The Paasche index: Calculates changes in real GDP using the final year prices
 - ▶ e.g., for year 2026 to 2027,

$$\text{Paasche index} = \frac{7650}{6500} = 1.177$$

- We can multiply the index by 100 so it is easier to read, but it is not required
- Over long-time intervals the two indexes can result in substantial differences

Chain Weighting

- The Fisher index (chain weighting) is the preferred approach to calculating real GDP
 - ▶ Geometric Average of the Laspeyres and Paasche index

$$\text{Fisher Index} = \sqrt{\text{Laspeyres} \times \text{Paasche}}$$

- ▶
- It is preferred because new goods are invented while others become obsolete, using only early or recent prices would be inaccurate.
- In the example above, we have the fisher index $= \sqrt{118.3 \times 117.7} = 118$ or in growth rate $118/100 - 1 = 18\%$

$$Y_{2026}^F \times 1.18 = 7650 \rightarrow Y_{2026}^F = 7650/1.18 = 6483$$

Chain Weighting

- To calculate the 2025 real GDP in chained prices, benchmarked to 2027, we have to calculate the following:

Let $G_{26,27}$ be the growth rate of the fisher index between year 26 to 27

$$Y_{2025}^F G_{25,26} \times G_{26,27} = Y_{2027}^F$$

We have calculate $G_{26,27}$ already, which is equal to 0.18

- Now we have to calculate $G_{25,26}$. From the table, we know that there is no real growth between these two years. Therefore,

$$G_{25,26} = \sqrt{\frac{6000}{6000} \times \frac{5000}{5000}} = 1$$

- Last, rearrange the equation above

$$Y_{2025}^F = \frac{Y_{2027}^F}{G_{25,26} \times G_{26,27}} = \frac{7650}{1.18} = 6483$$

This is the same as the number in year 2026

GDP Deflator

- Recall that

$$\text{Nominal GDP} = \text{Price level} \times \text{real GDP}$$

Rearrange that

$$\text{Price level} = \frac{\text{Nominal GDP}}{\text{real GDP}}$$

- This measure is called the GDP deflator
- Similar deflators exist for the various components of GDP, such as consumption and investment

Price Indexes and Inflation

- We can consider the following transformation:

$$\% \Delta \text{Nominal GDP} \approx \% \Delta \text{Price} + \% \Delta \text{real GDP}$$

This approximation hold if the change is relatively small

- If the $\% \Delta \text{Nominal GDP} = 27.5\%$ and the $\% \Delta \text{real GDP} = 18\%$, the percentage change in price is 9.5%
- The percentage change in price is also called inflation rate

Using Chain-Weighted Data

- Main motive for chain-weighted data: Rapid price changes in computer industry during the 1990s
 - ▶ The difference between the two indexes widen
- Main disadvantage:
 - ▶ The sum of real C, I, G, NX will not equal real chain-weighted GDP because the prices used in constructing the components are different
- General rule to follow:
 - ▶ For particular components of GDP, we look at the ratio of nominal variables
 - ▶ When you want real rates of economic growth, use the chain-weighted real measures

Comparing Economic Performance across Countries

- The exchange rate:
 - ▶ Represents the price at which various currencies are exchanged
- When comparing GDP across countries:
 - ▶ It is essential to express GDP in a common currency by adjusting it using the exchange rate
 - ▶ The resulting nominal GDP should then be further multiplied by the ratio of price levels in the respective countries
- World Bank and Penn World Tables are the two common source for exchange rate and price level adjusted data

An Example of Comparisons of Economies

Suppose we are trying to compare GDP in Canada and the United States.

- Use the exchange rate to turn Canadian dollars into U.S. dollars:

$$2.316 \text{ Trillion CAD} \times \frac{1}{1.327} = 1.745 \text{ Trillion USD}$$

- Adjust for relative price level of goods :

$$1.745 \text{ Trillion USD} \times \frac{1}{0.935} = 1.866 \text{ Trillion USD}$$

-

$$\begin{aligned} \text{Real GDP}_{CA}^{\text{US-price}} &= \text{price level}_{US} \times \text{real GDP}_{CA} = \text{price level}_{US} \times \frac{\text{nominal GDP}_{CA}}{\text{Price Level}_{CA}} \\ &= \frac{\text{price level}_{US}}{\text{Price Level}_{CA}} \times \text{nominal GDP}_{CA} \end{aligned}$$

The term $\frac{\text{price level}_{US}}{\text{Price Level}_{CA}} = \frac{1}{0.935}$ and nominal GDP_{CA} = 1.745 Trillion USD

Comparison of Countries

- In general, rich countries tend to have higher price levels than poor countries
- This is mainly because poor countries have lower wages

Next Lecture

- Next lecture will cover Ch.3